Student Performance Analytics System

Project Documentation

Python • Pandas • NumPy • CSV

Based on the supplied main.py source code

1. Project Overview

The Student Performance Analytics System is a Python program that processes marks for 10 students in HTML, CSS, JavaScript, SQL, and Python. It calculates each student's total and average, assigns a grade, determines pass/fail status, summarizes class performance, and saves the processed data to a CSV file.

2. Objectives

- Store and read student marks and attendance using CSV files.
- Calculate each student's total and average across five subjects.
- Assign a grade based on the calculated average.
- Determine pass/fail status using the minimum mark across the five subjects.
- Summarize class performance and export the processed report.

3. Technologies Used

- **Python** — main programming language.
- **Pandas** — creates, reads, processes, and exports tabular data.
- **NumPy** — rounds averages to two decimal places.
- **CSV** — stores input data and the generated performance report.

4. Project Files

File	Description
main.py	Python source code.
student_data.csv	Original student marks and attendance; created by the program.
student_performance.csv	Processed data with Total, Average, Grade, and Result columns.
README.md	Project description and run instructions.

5. Grade and Result Rules

Average	Grade
90 and above	S
80 to below 90	A
70 to below 80	B
60 to below 70	C
Below 60	F

A student is marked **Pass** if the minimum mark across HTML, CSS, JavaScript, SQL, and Python is at least 35. Otherwise, the result is **Fail**. Attendance is included in the data but is not used in grade or pass/fail calculations.

6. Input Dataset

The program contains 10 student records. Each record includes Student ID, Name, marks in five subjects, and Attendance.

ID	Name	HTML	CSS	JS	SQL	Python	Attend.
1	Anusha	87	87	89	85	78	92
2	Sriya	98	98	76	98	70	85
3	Sujitha	67	67	68	84	88	98
4	Sandhya	95	74	65	65	34	99
5	Ramya	87	82	73	88	54	67
6	Mounika	94	99	94	87	77	83
7	Hema	69	76	32	94	85	97
8	Shalini	88	87	98	97	99	55
9	Jyoshika	72	66	88	66	76	74
10	pavani	87	58	87	97	45	86

7. Implementation

1. Imports Pandas and NumPy.
2. Defines student IDs, names, marks, and attendance.
3. Writes input data to student_data.csv and reads it into a DataFrame.
4. Calculates each student's total and average for the five subjects.
5. Applies the grade function and pass/fail rule.
6. Prints the student report and class statistics.
7. Saves the processed DataFrame to student_performance.csv.

8. How to Run in VS Code

- Install Python and the Microsoft Python extension in VS Code.
- Open the folder containing main.py.
- Open Terminal → New Terminal.
- Install the required libraries if needed: `python -m pip install pandas numpy`
- Run the program: `python main.py`
- Check the terminal for the student report and class analysis. The program writes student_data.csv and student_performance.csv to the current working folder.

9. Program Output

Student Performance Report

ID	Name	Total	Average	Grade	Result
1	Anusha	426	85.20	A	Pass
2	Sriya	440	88.00	A	Pass
3	Sujitha	374	74.80	B	Pass
4	Sandhya	333	66.60	C	Fail
5	Ramya	384	76.80	B	Pass
6	Mounika	451	90.20	S	Pass
7	Hema	356	71.20	B	Fail
8	Shalini	469	93.80	S	Pass
9	Jyoshika	368	73.60	B	Pass
10	pavani	374	74.80	B	Pass

Class Analysis

Metric	Value
Class Average	79.50
Highest Average	93.80 (Shalini)
Lowest Average	66.60 (Sandhya)
Students Passed	8
Students Failed	2

10. Final Outcome

The program generates a structured student performance report and class-level summary. For the supplied dataset, the class average is **79.50**, the highest average is **93.80** (Shalini), the lowest average is **66.60** (Sandhya), and **8 students pass while 2 fail** under the current rule.

Code note: The supplied main.py needs a comma after the HTML marks list and before the CSS entry in the data dictionary. Correct this syntax issue before running the program.